

INSPECTION CERTIFICATE EN 10204-3.1 EN 10204:2005-01 D NOW L.P. 7402 N. Eldridge Parkway HOUSTON TX 77041 USA	Document No.:	465-1500733/001/P	Inspection No.:	Page: 1 / 3
	Purchase Order No.:	26673800	Manufacturer:	
	Benteler Order No.:	48010486	Manufacturer's brand:	
	Dispatch Note No.:		Stamp of the inspection representative:	
	Product:	SEAMLESS STEEL TUBES	Steelmaking process:	

Terms of delivery:	ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175/ISO 15156-1: 2020, ANSI/NACE MR0175/ISO 15156-2: 2020, ANSI/NACE MR0103/ISO 17945:2015
Dimensions-tolerances:	ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018
Steel grade:	GRADE B, GRADE C, GRADE BN, GRADE X42N
Delivery condition:	N
Product marking:	S: BENTELER - BENTELER-Sign - API 5L 1072 - API-Sign - Month/Year - A/SA 53 A/SA 106 NACE - 4,500" OD x 0,337" WT - B/C/BN/X42N - PSL2 - SMLS - Heat No. - Tube length feet - 2.970 - 15038034 - LONGHORN

AEZ = Etching ink marking FK = color marking FS = paint stenciling FSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PKE = tag marking PS = die stamp TS = Ink jet spray marking

Item	Dimensions	Heat No.	Test pressure	
			PSI	sec
0040	4 NPS - SCH 80 35,000 FT - 45,000 FT	319557	2970	5

Country of melt and pour

Melted & Manufactured in USA
Nucor, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0040	319557	0.160	0.267	1.00	0.012	0.002	0.11	0.02	0.07	0.17	0.003	0.001	0.002	0.0002

- Formula: $CE\ IIW = C + \frac{(Mn+6)}{6} + \frac{(Cr+Mo+V)}{5} + \frac{(Cu+Ni)}{15} \leq 0,43\ \%$
- Formula: $Cr+Cu+Mo+Ni+V \leq 1.00\ \%$
- Formula: $Nb+V \leq 0.06\ \%$
- Formula: $Nb+Ti+V \leq 0.15\ \%$

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Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0040	319557	0,369	0,373	0,004	0,006

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0040	319557	0.180	0.280	1.00	0.011	0.001	0.11	0.02	0.07	0.17	0.003	0.001	0.003	0.0001
0040	319557	0.180	0.280	1.00	0.012	0.001	0.11	0.02	0.07	0.17	0.003	0.001	0.003	0.0001

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr+Mo+V)/5) + ((Cu+Ni)/15) \leq 0,43 \%$

2. Formula: $Cr+Cu+Mo+Ni+V \leq 1.00 \%$

3. Formula: $Nb+V \leq 0.06 \%$

4. Formula: $Nb+Ti+V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0040	319557	0,389	0,373	0,004	0,007
0040	319557	0,389	0,373	0,004	0,007

Test results

The tubes have been submitted to a leak tightness test by:	Hydrostatic test: acc. to API 5L; Test pressure/Time-record, the test records/charts shall be available for examination by the inspector	PASSED
The tubes are non destructive tested:	ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0,125 IN.; ET - Notch shape/hole: drilled hole	PASSED
Visual inspection:	PASSED	Dimensions examination: PASSED
		Flattening test: PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0,5/Rm
Requirements			IN	RT 0,5 % PSI 42100-71800	Rm PSI 70000-95000	A2" %		MAX 0,93
0040	000011	319557	1,002 x 0,346	49023	76435	38,00	14,00	0,64

Elongation formula

1. Formula $AF = 625000 * AXC^{0,2} / U^{0,9}$

Notched bar impact test [1 CHARPY_V]

Specimen position: longitudinal (L); Test temperature: -50 °F

Item	Specimen No.	Heat No.	Specimen dimensions			Absorbed energy		Impact strength	Shear fracture	
Requirements			Length	Width	Height	single	average		single	average
			mm	mm	mm	ft-lbf	ft-lbf		%	%
			55	6,67	10,00					
0040	000011	319557	55	6,67	10,00	15			10	
				6,67	10,00	35			10	
				6,67	10,00	24			10	10

Protection

outside surface varnished

Item	Heat No.	Production No.
0040	319557	15038034

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156: HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945: HRC max. 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Data converted from SI units, No weld repair has been carried out;

Certificate-Remark: LONGHORN